

AMENDMENT NO. 2 DECEMBER 1986

TO

IS:6848-1979 SPECIFICATION FOR LEAD-ACID BATTERIES
FOR TRAIN LIGHTING AND AIRCONDITIONING SERVICES

(First Revision)

(Page 22, Appendix F) - Substitute the following
for the existing Appendix:

A P P E N D I X F

(Fig. 2A, 2B and 3)

DETERMINATION OF AVERAGE THICKNESS OF LEAD-COATING
(MODIFIED PROCEDURE FOR APPLICABILITY TO COPPER,
AS ALUMINIUM CONDUCTOR CONNECTORS)

F-1. The lead plating on the specimen shall be dissolved in a mixture of 80 parts of glacial acetic acid, 10 parts of hydrogen peroxide, 30 percent and 10 parts of distilled water by volume at a temperature of 60° approx. till the base material is observed visually.

F-2. Remove the specimen and wash it with distilled water. Collect the washings. To the solution of hydrogen peroxide, glacial acetic acid and add dilute sulphuric acid (sp gr 1.200 approx) till lead (Pb) is completely precipitated as $PbSO_4$.

F-3. Allow the precipitate to stand at room temperature to settle down. Decant the supernatant liquid and then filter the precipitate through a previously weighed Whatman filter paper No. 1. Wash the precipitate with water containing (10 percent by volume) H_2SO_4 until free of soluble impurities and then with 50 percent alcohol solution to remove the free acid. Dry the precipitate to a constant weight and determine the average thickness as follows:

$$\begin{aligned}
 \text{a) Average lead plating thickness (in cm)} & \left\{ \begin{array}{l} \text{Weight of ppt. of PbSO}_4 \text{ (gm)} \\ \text{11.34 (Density of lead)} \end{array} \right\} = \frac{0.6833 \text{ conversion factor}}{\text{Surface area of the specimen in cm}^2} \\
 \\
 \text{b) Average lead-plating thickness in microns} & \left\{ \begin{array}{l} \text{Average Lead Plating Thickness in cm} \end{array} \right\} = \text{Average Lead Plating Thickness in cm} \times 10
 \end{aligned}$$